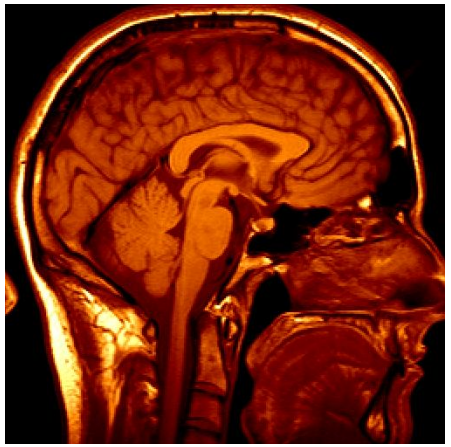
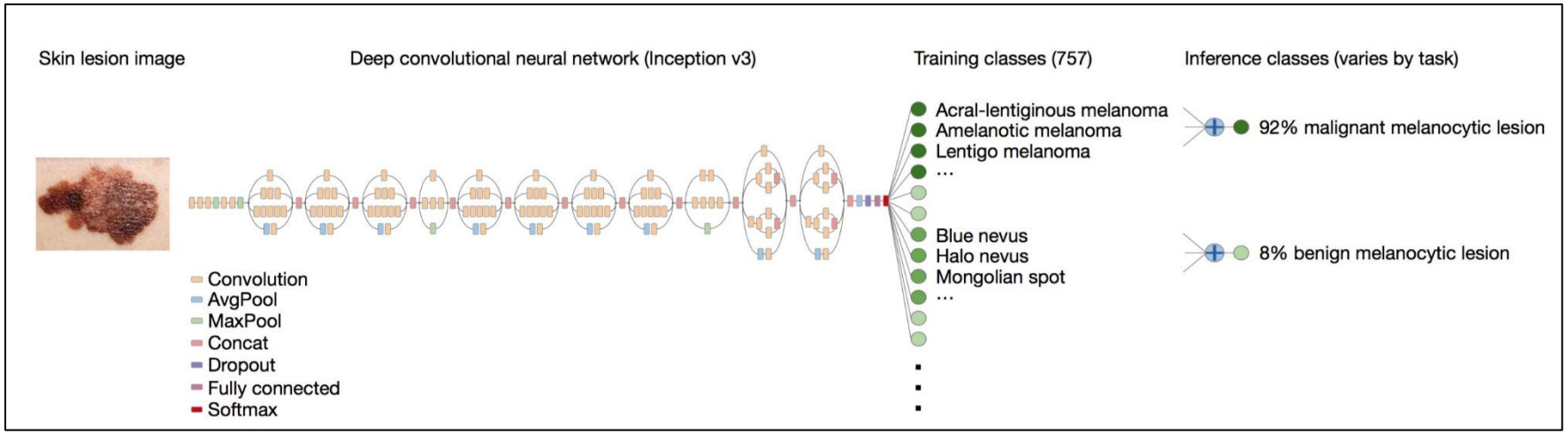
Medical imaging

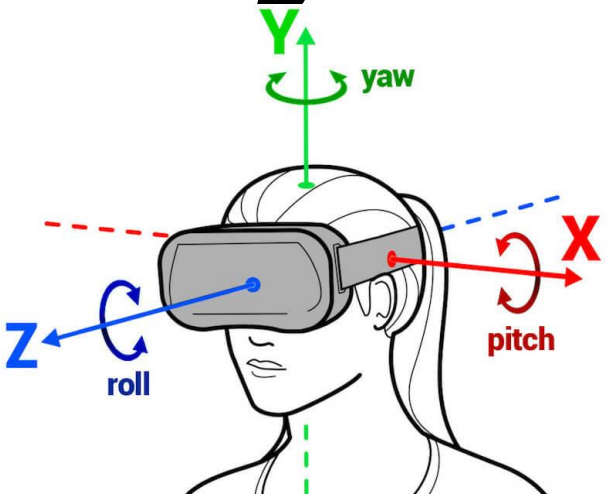


3D imaging
(MRI, CT)



Skin cancer classification with deep learning
https://cs.stanford.edu/people/esteva/nature/

Virtual & Augmented Reality



6DoF head tracking



Hand & body tracking



3D scene understanding



3D-360 video capture

Current state of the art

- You just saw many examples of current systems.
  - Many of these are less than 10 years old
- Computer vision is an active research area, and rapidly changing
  - Many new apps in the next 5 years
  - Deep learning and generative methods powering many modern applications
- Many startups across a dizzying array of areas
  - Generative AI, robotics, autonomous vehicles, medical imaging, construction, inspection, VR/AR, …

Why is computer vision difficult?



Viewpoint variation



Illumination



Scale

Why is computer vision difficult?



Intra-class variation



Motion (Source: S. Lazebnik)



Background clutter

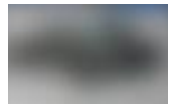


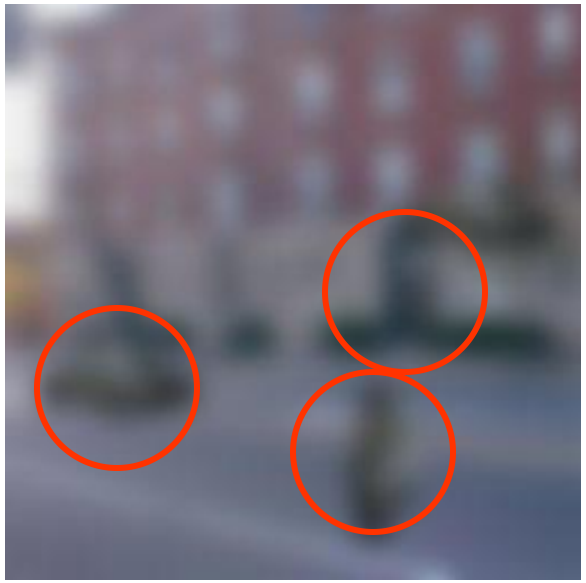
Occlusion

Challenges: local ambiguity











slide credit: Fei-Fei, Fergus & Torralba

But there are lots of visual cues we can use…

香港城市大學（東莞）
City University of Hong Kong
(Dongguan)



Source: S. Lazebnik

CS5187: Vision and Image

- Project-based course whose goal is to teach you the fundamentals of computer vision – image processing, geometry, recognition – in a hands-on way

- This course covers **fundamentals**, **including mathematical fundamentals**. It is not a course specifically on deep learning or generative AI, though those topics will be covered.